Science Notebook

CHATGPT EDITION

Evidence before energy

SUPERVISED-ONLY PROJECT

One named, sober adult controls the launcher, fuel, ignition, commands, misfires, and cease-fire. Minors do not fuel, ignite, clear a misfire, or handle a loaded launcher. This set is educational guidance, not a certification of any component or assembly.

THE LEARNING SEQUENCE

Complete the paper models and cold measurements first. Live operation is optional, adult-controlled, outdoors only, and permitted only after every gate on the **Range Command Card** passes. Never change fuel, projectile fit, angle, or hardware merely to chase a larger result.

Model 1 · A bounded flame

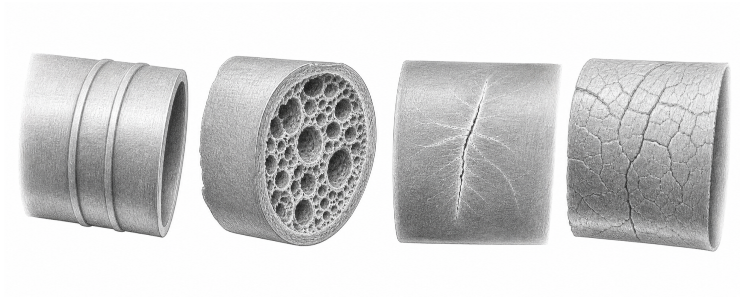
Combustion is possible only inside a fuel's flammable range. Below its lower limit the mixture is too lean; above its upper limit it is too rich. That makes "more fuel" scientifically wrong as a general rule. Product formulation, sprayer, temperature, chamber history and mixing all vary, so this edition provides no universal dose.

Too lean · bounded region · too rich. The drawing is conceptual, not a dose scale.

Write before observing	
Claim	A bounded mixture can ignite; mixtures outside the bounds may not.
Prediction	Adding fuel can move a mixture toward *or away from* the flammable region.
Safety consequence	Never top up a misfire; hold 60 seconds, vent fully, and return to the cold state. Never use liquid accelerants or oxygen enrichment.

Model 2 · Pressure is not approval

Heating gas in a constrained volume can raise pressure, but a plumbing pressure mark describes a listed plumbing product under specified conditions. It does not certify a homemade combustion assembly. Use only solid-wall, pressure-marked pipe *and* pressure fittings; never DWV/cellular core and never compressed air or gas. Temperature, sunlight, impact, workmanship and chemical exposure can reduce margin.



Sound surface · cellular core · stress crack · crazing. A mark cannot cancel a defect.

Material audit

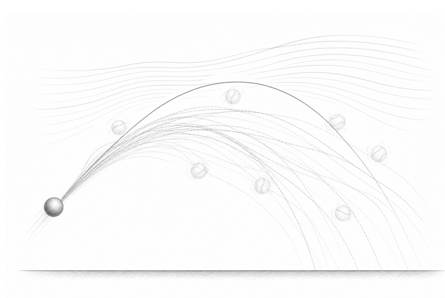
Evidence read directly from every part	Pass/fail	Adult initials
Solid wall; pressure standard/rating; matching pressure fitting		
No DWV, cellular/foam core, cracks, whitening, crazing or deformation		
Cement/primer compatible; batch usable; label instructions retained		
Cure start recorded; 48 h minimum and manufacturer longer time satisfied		

Model 3 · A useful trajectory, with a boundary

For launch speed v, angle θ, and gravity g, the no-drag model is

$$x(t) = v\cos(\theta)t, \qquad y(t) = v\sin(\theta)t - \tfrac{1}{2}gt^2.$$

It explains the parabolic ideal and separates horizontal from vertical motion. It does not include drag, potato shape, wind, launch-height difference, tumble, fragmentation, or shot-to-shot variation.



One ideal path versus a family of plausible outcomes under omitted effects.

Evidence table — observations, not performance tuning

Trial	Cold/live	Predicted observation	Observed / uncertainty
1			
2			
3			

Model 4 · Controls beat measurement

Impulse sound can injure immediately, and ordinary meters may clip the peak. Therefore a phone reading cannot clear the hazard. Distance, limited attendance, earplugs plus earmuffs, impact-rated eye protection, and a disciplined firing line remain required regardless of the displayed number.

Layered controls remain necessary when a phone waveform clips or misses the peak.

A good conclusion

State what the model explained, what the evidence showed, and what remains unknown. The strongest result is often a justified stop decision. Do not report a PVC pressure rating as a combustion rating, a computed range as containment, or one successful firing as proof of safety.